



**BHARATI VIDYAPEETH'S
INSTITUTE OF COMPUTER APPLICATIONS & MANAGEMENT**

(Affiliated to Guru Gobind Singh Indraprastha University,
Approved by AICTE, New Delhi)

Computer Networks Lab

(MCA-161)

Practical File

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MCA 1st Sem, Sec B

Exp. No.	Description of Lab Exercise	BTL	CO	Date	Page No.	Sign. Of teacher
1.	Understand the various Devices used in Computer networks.	BTL2	CO1	3.11.25		
2.	Run the basic networking commands in command prompt and give the results as per your system. The commands are: a) IPCONFIG b) IPCONFIG/ALL c) NSLOOKUP d) PING e) Traceroute	BTL2, BTL4	CO1	3.11.25		
3	a) Learn Cisco-packet tracer and connect two PC using Ethernet cross over cable. b) connect 2 computers without a Connecting Device, give them a IP address, check all the configurations of both computer in its command prompt	BTL2, BTL3	CO2, CO3	3.11.25		
4	Connect 6 computers with a Hub.	BTL2, BTL6	CO3	11.11.25		
5	a) Connect 6 PCs through a Switch b) Differentiate between switch and Hub	BTL2, BTL6	CO3	11.11.25		
6	Demonstrate classful addressing through 2940 switch and ethernet straight through cable. a) Create one LAN using a switch and 3 PC's. Give 2 PC's IP address as 10.0.0.1 and 10.0.0.2, third PC IP address is 20.0.0.3. What is happening when sending a message from	BTL5, BTL6	CO5, CO6	11.11.25		

	PC1 to PC3 explain. b) When you assign an IP address as 10.0.0.0 and 10.255.255.255, what happen. c) In the same LAN created in part 'a', change the IP address of PC 3 to 199.166.2.0, now send a message from PC1 to PC3 explain the scenario.					
7	Install three Workstation or Computers in different networks with different IP address and connect them through a router.	BTL2, BTL3	C03	17.11.25		
8	Demonstrate dynamic routing using RIP protocol by connecting with 2 Switches, 6 PCs and 2 Routers.	BTL 5, BTL6	C05, C06	17.11.25		
9	Install one repeater to connect two networks in different buildings and establish communication between the workstation.	BTL2, BTL4	C04	17.11.25		

Problem 1

Various devices on computer networks

1 .Cables

cables play a critical role in connecting various devices and enabling communication between them. Here are the main types of cables used in computer networks:

1. Ethernet Cables

Cat5e (Category 5e)

Speed: Up to 1 Gbps (Gigabit Ethernet)

Distance: Up to 100 meters

Standard: 1000Base-T

Cat6 (Category 6)

Speed: Up to 10 Gbps (short distances, up to 55 meters)

Distance: Up to 100 meters at 1 Gbps

Standard: 10GBase-T (short-range)

Cat6a (Category 6a)

Speed: Up to 10 Gbps

Distance: Up to 100 meters

Standard: 10GBase-T

Cat7 (Category 7)

Speed: Up to 10 Gbps

Distance: Up to 100 meters

Standard: 10GBase-T

Shielding: Additional shielding for reduced interference

Cat8 (Category 8)

Speed: Up to 25-40 Gbps

Distance: Up to 30 meters

Standard: 25GBase-T / 40GBase-T

2. Coaxial Cables

10Base2

Speed: 10 Mbps

Distance: Up to 200 meters

Standard: 10Base2 (Thin Ethernet)

10Base5

Speed: 10 Mbps

Distance: Up to 500 meters

Standard: 10Base5 (Thick Ethernet, less common today)

3. Fiber Optic Cables

Single-Mode Fiber (SMF)

Speed: Varies; typically up to 100 Gbps

Distance: Long distances (several kilometers)

Standard: 100GBase-LR, 100GBase-ER, etc.

Multi-Mode Fiber (MMF)

Speed: Up to 40 Gbps or 100 Gbps (shorter distances)

Distance: Up to 550 meters (for OM4 fiber)

Standard: 10GBase-SR, 40GBase-SR4, etc.

4. Direct Attach Copper (DAC)

Speed: Up to 10 Gbps, 25 Gbps, 40 Gbps, etc.

Distance: Short-range (typically up to 7 meters)

Standard: 10GBase-CU, 25GBase-CR, 40GBase-CR4

5. Active Optical Cables (AOC)

Speed: Up to 10 Gbps, 25 Gbps, 40 Gbps, etc.

Distance: Short to medium range (varies, often up to 100 meters)

Standard: 10GBase-SR, 25GBase-SR, 40GBase-SR4

2.Repeater

Repeaters are devices used in network communication to extend the distance of a signal by amplifying or regenerating it. Boosts signal strength to overcome distance limitations and restore signal quality.

3.Hubs

Hubs are basic network devices used to connect multiple devices in a network, allowing them to communicate with each other. Distributes incoming data packets to all connected devices (ports) in the network.

4.Switches

Connects multiple devices within a local area network (LAN) and manages data traffic efficiently. Uses MAC addresses to forward data only to the specific device (port) intended for that data, reducing collisions and improving network efficiency.

5.Bridges

Connects and filters traffic between two or more network segments, making them function as a single network. Uses MAC addresses to determine if a data packet should be forwarded or filtered, reducing network traffic.

6.Router

Connects different networks (e.g., a LAN to the internet) and routes data packets between them. Uses IP addresses to determine the best path for forwarding packets across networks.

7.Gateway

Acts as a point of connection between different networks or network segments, often translating between different network protocols or systems. Routes data between networks that use different protocols (e.g., between a local network and the internet) and can perform tasks like protocol conversion, network address translation (NAT), and firewall protection.

8.Modem

Modulates and demodulates signals to enable digital data transmission over analog communication lines, such as telephone lines or cable systems. Converts digital data from a computer into an analog signal for transmission over communication lines and vice versa.

9.Intrusion Detection System (IDS)

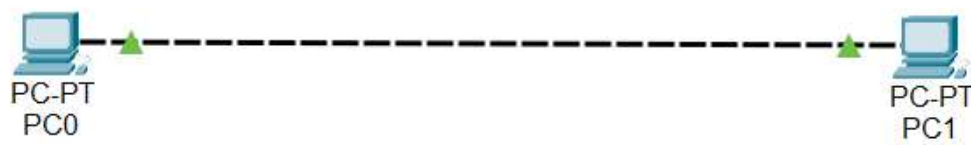
Monitors network or system activity for signs of suspicious or malicious behavior and alerts administrators about potential threats. Analyzes traffic or system logs to detect patterns or anomalies that may indicate an intrusion or attack.

10.Firewall

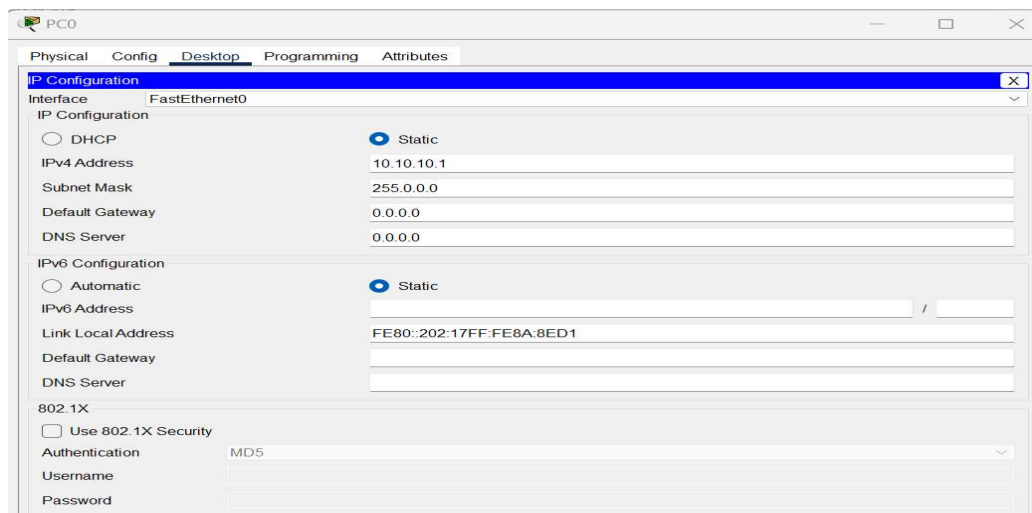
Controls and filters incoming and outgoing network traffic based on predetermined security rules to prevent unauthorized access and attacks. Inspects packets of data and enforces rules to allow or block traffic between networks or between devices and networks.

Problem 2.

- a) Learn Cisco-packet tracer and connect two PCs using Ethernet cross over cable.



- b) Connect 2 computers without a Connecting Device, give them a IP address, check all the configurations of both computer in its command prompt.



PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.10.2

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:8FFF:FED4:75B8

Default Gateway

DNS Server

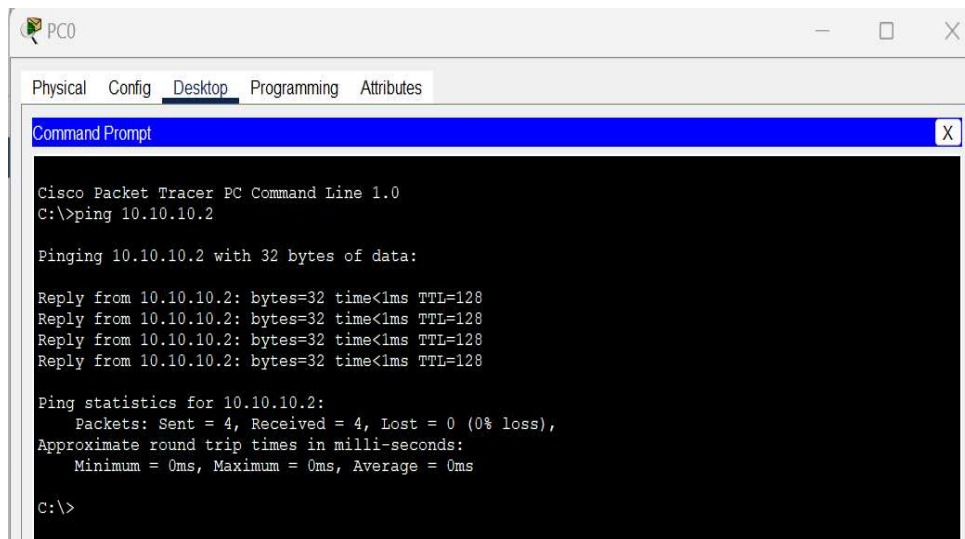
802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password



```

PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

```

C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 10.10.10.1: bytes=32 time<1ms TTL=128
Reply from 10.10.10.1: bytes=32 time<1ms TTL=128
Reply from 10.10.10.1: bytes=32 time<1ms TTL=128
Reply from 10.10.10.1: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC0	ICMP		0.000	N	1	(edit)	(delete)

Problem 3:

Run the basic networking commands in command prompt and give the results as per your system. The commands are:

a) IPCONFIG

```
(c) Microsoft Corporation. All rights reserved.  
C:\Users\LENOVO>ipconfig  
  
Windows IP Configuration  
  
Ethernet adapter Ethernet 2:  
  
    Connection-specific DNS Suffix  . :  
    Link-local IPv6 Address . . . . . : fe80::445:55d:d3c8:8de3%19  
    IPv4 Address. . . . . : 192.168.56.1  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . :  
  
Wireless LAN adapter Local Area Connection* 1:  
  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix  . :  
  
Wireless LAN adapter Local Area Connection* 2:  
  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix  . :  
  
Wireless LAN adapter Wi-Fi:  
  
    Connection-specific DNS Suffix  . : bbrouter  
    Link-local IPv6 Address . . . . . : fe80::8659:d286:bdf4:cc88%18  
    IPv4 Address. . . . . : 192.168.1.4  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . : 192.168.1.1  
  
Ethernet adapter Bluetooth Network Connection:  
  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix  . :
```

b) IPCONFIG/ALL

```

C:\Users\LENOVO>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : LAPTOP-NGLFE04F
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : bbrouter

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix . :
    Description . . . . . : VirtualBox Host-Only Ethernet Adapter
    Physical Address. . . . . : 0A-00-27-00-00-13
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-Local IPv6 Address . . . . : fe80::445:55d:d3c8:8de3%19(Preferred)
    IPv4 Address. . . . . : 192.168.56.1(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
    DHCPv6 IAID . . . . . : 503971879
    DHCPv6 Client DUID. . . . . : 00-01-00-01-30-31-E0-5F-D4-AE-52-5B-96-1D
    NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
    Physical Address. . . . . : 9C-65-EB-7A-41-27
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

```

```

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
    Physical Address. . . . . : 9E-65-EB-7A-41-26
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : bbrouter
    Description . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
    Physical Address. . . . . : 9C-65-EB-7A-41-26
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-Local IPv6 Address . . . . : fe80::8659:d286:bdf4:cc88%18(Preferred)
    IPv4 Address. . . . . : 192.168.1.4(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : 19 November 2025 19:27:36
    Lease Expires . . . . . : 24 November 2025 10:11:38
    Default Gateway . . . . . : 192.168.1.1
    DHCP Server . . . . . : 192.168.1.1
    DHCPv6 IAID . . . . . : 295462379
    DHCPv6 Client DUID. . . . . : 00-01-00-01-30-31-E0-5F-D4-AE-52-5B-96-1D
    DNS Servers . . . . . : 192.168.1.1
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Bluetooth Device (Personal Area Network)
    Physical Address. . . . . : 9C-65-EB-7A-41-2A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

```

c) NSLOOKUP

```
PS C:\Users\LENOVO> nslookup www.bvicam.in
Server:   RTK_GW.bbrouter
Address:  192.168.1.1

Non-authoritative answer:
Name:     bvicam.in
Address:  14.140.205.246
Aliases:  www.bvicam.in
```

d) PING

```
PS C:\Users\LENOVO> ping www.google.com

Pinging www.google.com [142.250.193.36] with 32 bytes of data:
Reply from 142.250.193.36: bytes=32 time=3ms TTL=118
Reply from 142.250.193.36: bytes=32 time=18ms TTL=118
Reply from 142.250.193.36: bytes=32 time=38ms TTL=118
Reply from 142.250.193.36: bytes=32 time=50ms TTL=118

Ping statistics for 142.250.193.36:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 50ms, Average = 27ms
PS C:\Users\LENOVO>
```

e) Traceroute

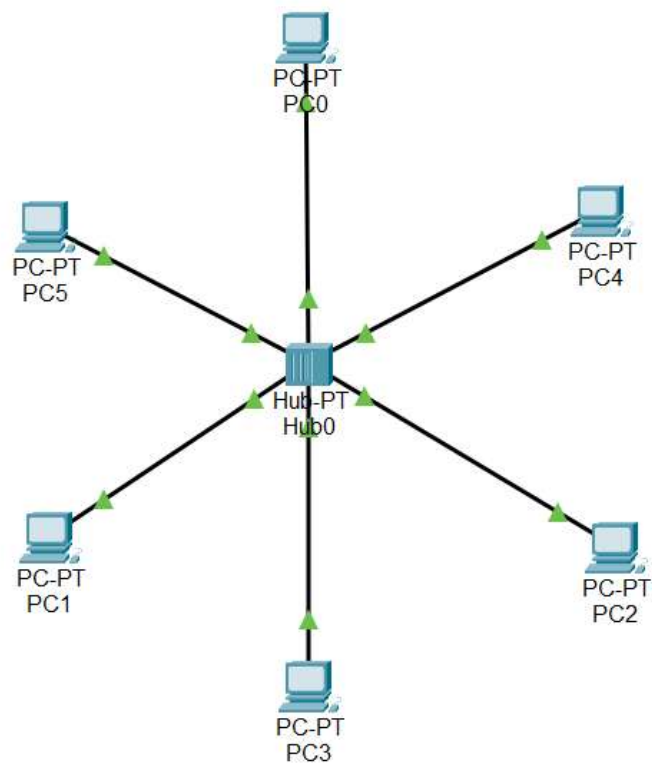
```
PS C:\Users\LENOVO> tracert www.google.com

Tracing route to www.google.com [142.250.193.36]
over a maximum of 30 hops:

  1     2 ms    <1 ms    <1 ms  RTK_GW.bbrouter [192.168.1.1]
  2     3 ms     2 ms     2 ms  252.6.27.1.hathway.com [27.6.252.1]
  3      *      *      *      Request timed out.
  4      *      *      *      Request timed out.
  5      *      *      *      Request timed out.
  6      *      *      *      Request timed out.
  7      *      *      *      Request timed out.
  8     3 ms     8 ms     3 ms  del11s15-in-f4.1e100.net [142.250.193.36]

Trace complete.
```

Problem 4: Connect 6 computers with a Hub.



PC0

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.1

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::210:11FF:FE84:AA3

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.2

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:BCFF:FE51:2788

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.3

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::240:BFF:FE9C:70C4

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.4

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20A:F3FF:FEA7:2916

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC4

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.5

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:47FF:FE96:41AA

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC5

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.6

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:17FF:FE9D:3A95

Default Gateway

DNS Server

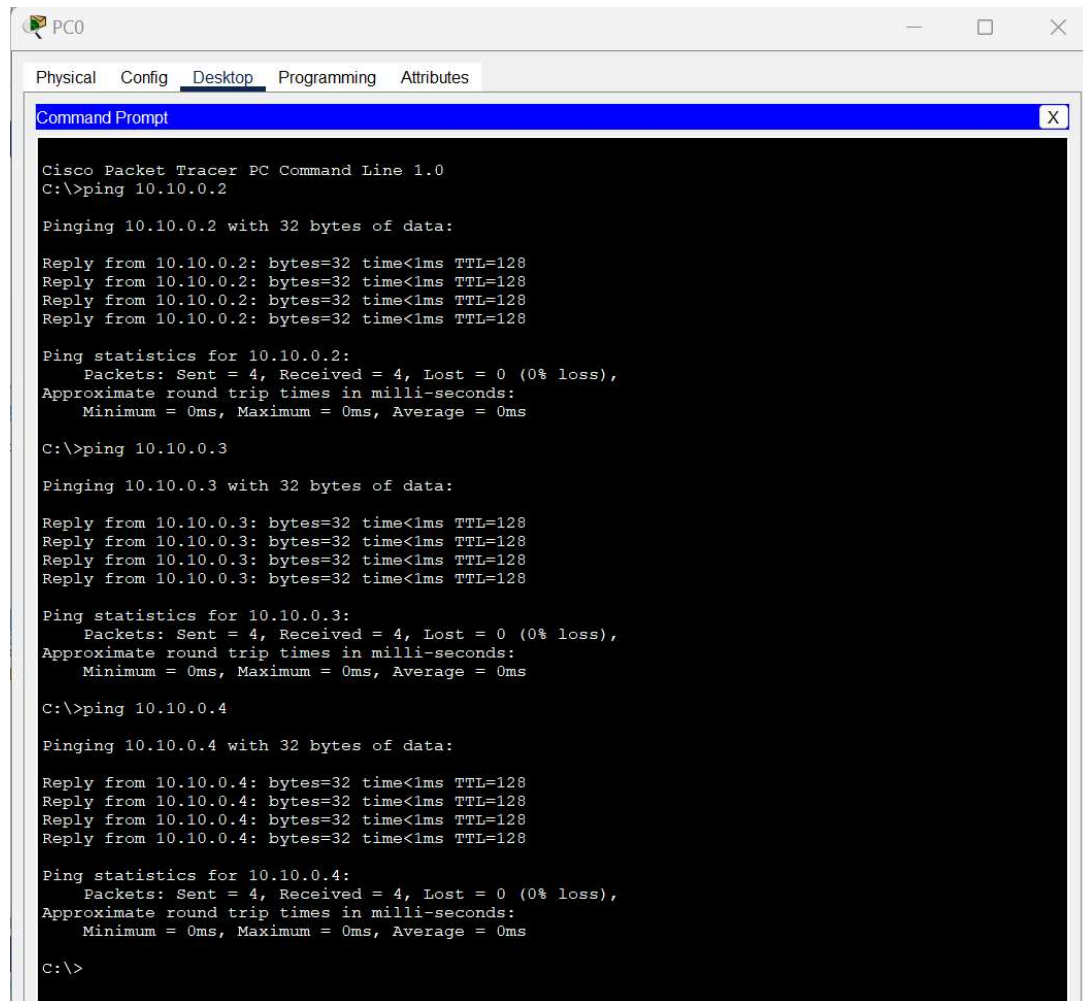
802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.0.2

Pinging 10.10.0.2 with 32 bytes of data:

Reply from 10.10.0.2: bytes=32 time<1ms TTL=128
Reply from 10.10.0.2: bytes=32 time<1ms TTL=128
Reply from 10.10.0.2: bytes=32 time<1ms TTL=128
Reply from 10.10.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.0.3

Pinging 10.10.0.3 with 32 bytes of data:

Reply from 10.10.0.3: bytes=32 time<1ms TTL=128
Reply from 10.10.0.3: bytes=32 time<1ms TTL=128
Reply from 10.10.0.3: bytes=32 time<1ms TTL=128
Reply from 10.10.0.3: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.0.4

Pinging 10.10.0.4 with 32 bytes of data:

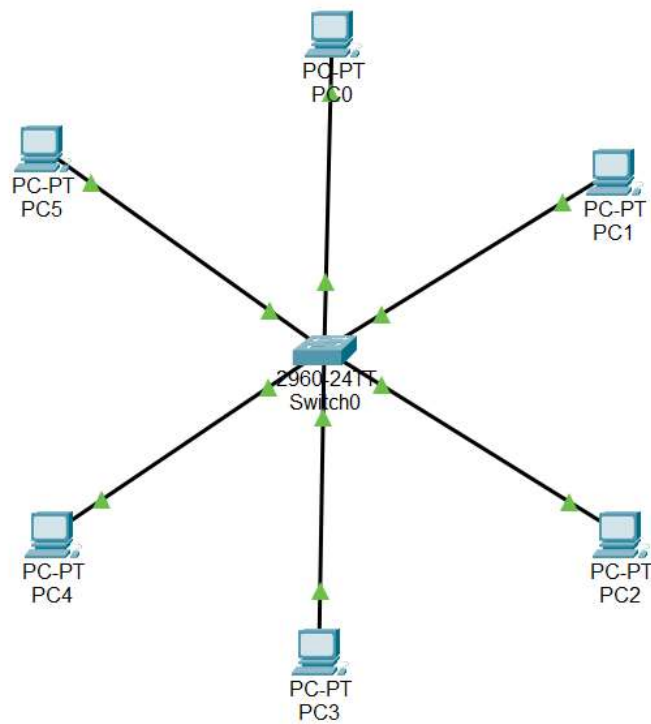
Reply from 10.10.0.4: bytes=32 time<1ms TTL=128
Reply from 10.10.0.4: bytes=32 time<1ms TTL=128
Reply from 10.10.0.4: bytes=32 time<1ms TTL=128
Reply from 10.10.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Problem 5:

a) Connect 6 PCs through a Switch



PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.1

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::210:11FF:FE84:AA3

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.2

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:BCFF:FE51:2788

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC2

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.3

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::240:BFF:FE9C:70C4

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC3

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.4

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20A:F3FF:FEA7:2916

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC4

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.5

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:47FF:FE96:41AA

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC5

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.0.6

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:17FF:FE9D:3A95

Default Gateway

DNS Server

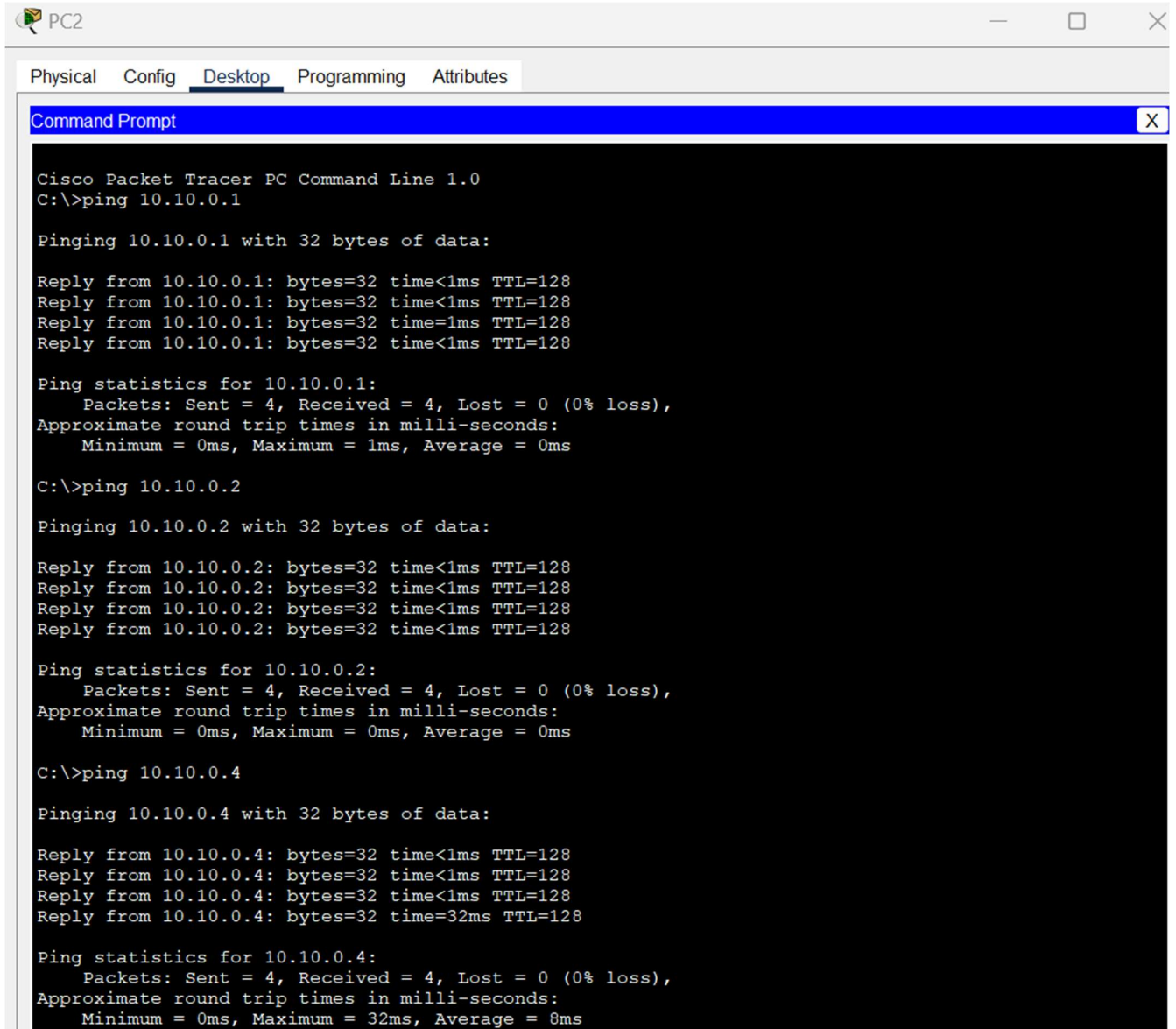
802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password



```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.0.1

Pinging 10.10.0.1 with 32 bytes of data:

Reply from 10.10.0.1: bytes=32 time<1ms TTL=128
Reply from 10.10.0.1: bytes=32 time<1ms TTL=128
Reply from 10.10.0.1: bytes=32 time=1ms TTL=128
Reply from 10.10.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.0.2

Pinging 10.10.0.2 with 32 bytes of data:

Reply from 10.10.0.2: bytes=32 time<1ms TTL=128
Reply from 10.10.0.2: bytes=32 time<1ms TTL=128
Reply from 10.10.0.2: bytes=32 time<1ms TTL=128
Reply from 10.10.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.0.4

Pinging 10.10.0.4 with 32 bytes of data:

Reply from 10.10.0.4: bytes=32 time<1ms TTL=128
Reply from 10.10.0.4: bytes=32 time<1ms TTL=128
Reply from 10.10.0.4: bytes=32 time<1ms TTL=128
Reply from 10.10.0.4: bytes=32 time=32ms TTL=128

Ping statistics for 10.10.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 32ms, Average = 8ms
```

b) differentiate between switch and Hub

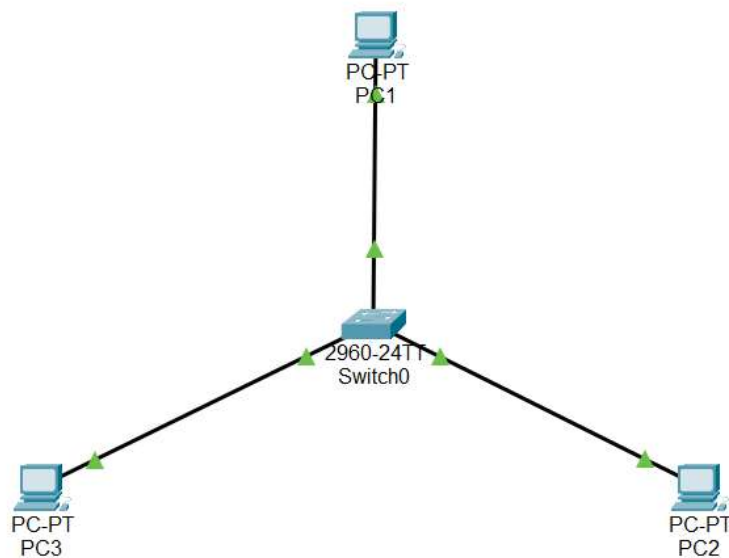
Hub: Broadcasts data to all connected devices, leading to potential collisions.

Switch: Sends data directly to the destination device based on MAC addresses, improving efficiency.

Problem 6:

Demonstrate classful addressing through 2940 switch and ethernet straight through cable.

- a) Create one Lan using a switch and 3 PC's. Give 2 PC's IP address as 10.0.0.1 and 10.0.0.2, third PC IP address is 20.0.0.3. What is happening when sending a message from PC1 to PC3 explain.



PC1

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address10.0.0.1

Subnet Mask255.0.0.0

Default Gateway0.0.0.0

DNS Server0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local AddressFE80::201:97FF:FED8:877

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

PC2

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address10.0.0.2

Subnet Mask255.0.0.0

Default Gateway0.0.0.0

DNS Server0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local AddressFE80::201:C7FF:FE83:74C0

Default Gateway

DNS Server

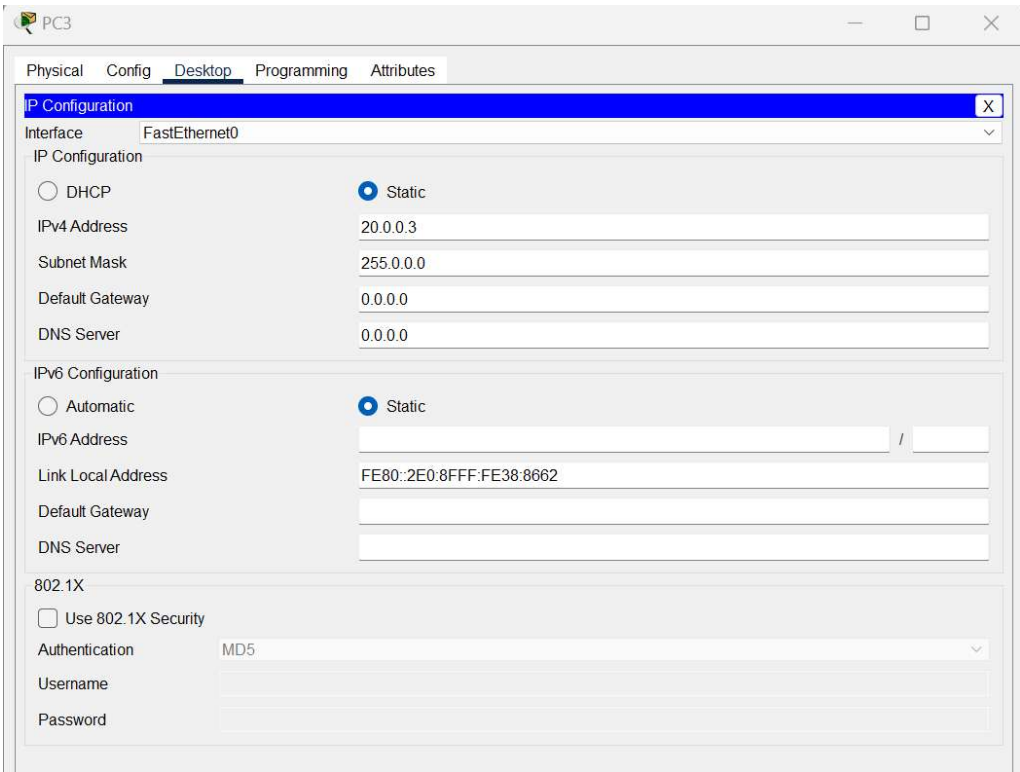
802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password



```

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=15ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 3ms

C:\>ping 20.0.0.3

Pinging 20.0.0.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 20.0.0.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Failed	PC1	PC3	ICMP		0.000	N	1	(edit)	(delete)

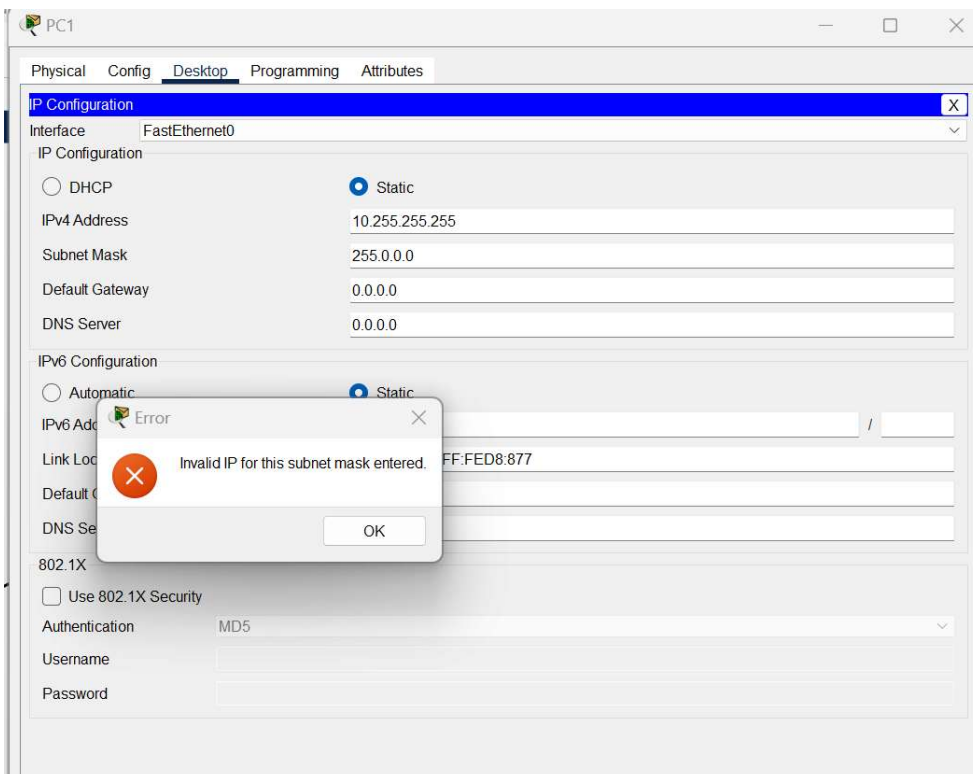
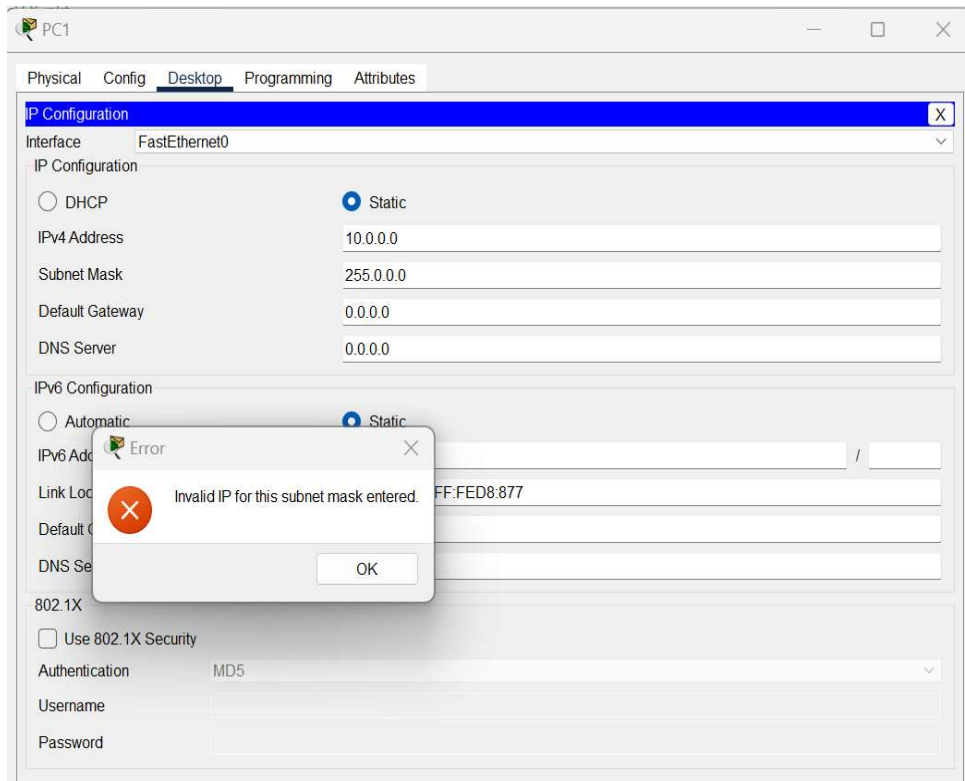
- The addresses **10.0.0.1** and **10.0.0.2** are in the same network (network address 10.0.0.0).
- The address **20.0.0.3** is in a different network (network address 20.0.0.0) since it begins with a different first octet.

When **PC1** tries to send a message to **PC3**, it checks if PC3's IP address (20.0.0.3) is in the same network. Using the subnet mask 255.0.0.0, PC1 will determine that:

- **10.0.0.x** (PC1 and PC2's addresses) belong to the **10.0.0.0 network**.
- **20.0.0.3** (PC3's address) belongs to a different **20.0.0.0 network**.

Since PC1 and PC3 are on **different networks**, they can't communicate directly using the switch, which operates only within the same network. As a result, the message from PC1 to PC3 will **not be delivered** because the switch doesn't perform routing between networks.

b) When you assign an IP address as 10.0.0.0 and 10.255.255.255, what happen.



- **10.0.0.0**: This IP is reserved as the network address, so it **cannot be assigned** to a device. Attempting to do so will result in an error because the network address is used to identify the network, not a specific device.
- **10.255.255.255**: This IP is reserved as the broadcast address, so it **cannot be assigned** to a device either. This address is used to broadcast messages to all devices on the 10.0.0.0 network.

c) In the same LAN created in part 'a', change the IP address of PC 3 to 199.166.2.0, now send a message from PC1 to PC3 explain the scenario.

```
C:\>ping 199.166.2.0

Pinging 199.166.2.0 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 199.166.2.0:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC1	PC3	ICMP		0.000	N	0	(edit)	(delete)

When PC1 tries to communicate with PC3, it sees that **199.166.2.0** does not fall within its own network (10.0.0.0/8), as defined by the subnet mask 255.0.0.0.

Since **PC1 and PC3 are now in different IP networks** (10.0.0.0 for PC1 and 199.166.2.0 for PC3), they cannot communicate directly through the switch.

The switch only allows communication within the same network, and it does not have the capability to route packets between different networks.

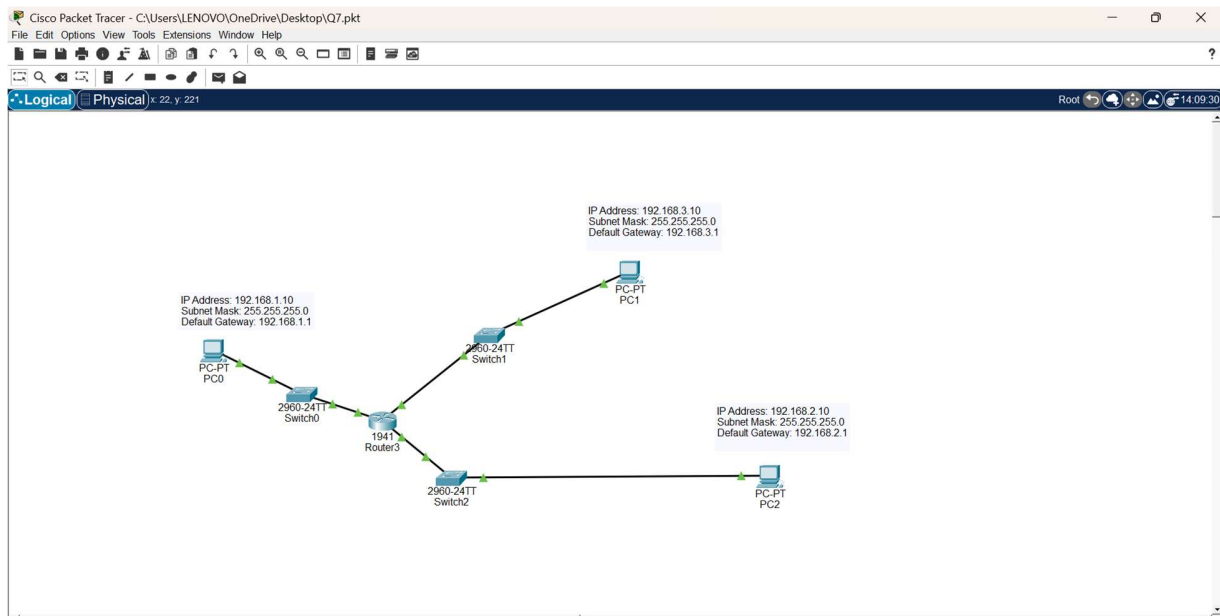
The message from PC1 to PC3 will not be delivered.

A **router** would be needed in this case to enable communication between different networks.

Without a router, PCs in different networks cannot communicate directly, even if they are connected to the same switch.

Problem 7:

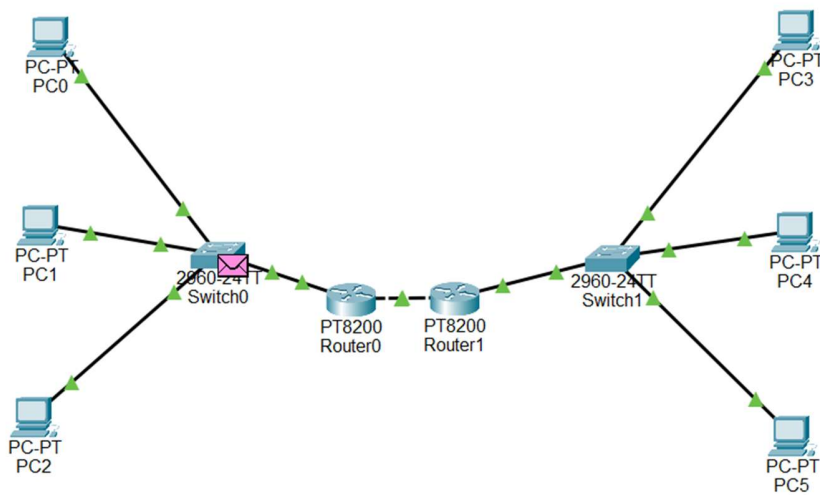
Install three Workstation or Computers in different networks with different IP address and connect them through a router..



..

Problem 8:

Demonstrate dynamic routing using RIP protocol by connecting with 2 Switches, 6 PCs and 2 Routers.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

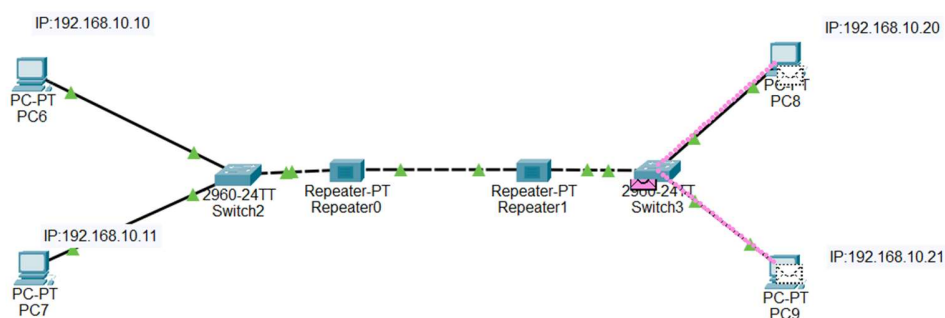
Request timed out.
Reply from 192.168.2.10: bytes=32 time=10ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

Problem 9:

Install one repeater to connect two networks in different buildings and establish communication between the workstation.



```
PC6
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```